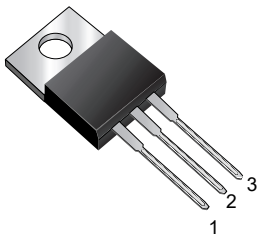
Dual Common Cathode Schottky Rectifier

TO-220AB





PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	2 x 12.5 A
V_{RRM}	35 V, 45 V
I_{FSM}	150 A
V_F	0.73 V at 30 A
T_J max.	150 °C
Package	TO-220AB
Diode variation	Common cathode

FEATURES

- Power pack
- Guardring for overvoltage protection
- Lower power losses, high efficiency
- Low forward voltage drop
- High forward surge capability
- High frequency operation
- Solder bath temperature 275 °C maximum, 10 s, per JESD 22-B106
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

TYPICAL APPLICATIONS

For use in low voltage, high frequency rectifier of switching mode power supplies, freewheeling diodes, DC/DC converters, or polarity protection application.

MECHANICAL DATA

Case: TO-220AB
Epoxy meets UL 94 V-0 flammability rating
Base P/N-E3 - RoHS-compliant, commercial grade

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD22-B102
E3 suffix meets JESD 201 class 1A whisker test

Polarity: as marked

Mounting Torque: 10 in-lbs maximum

MAXIMUM RATINGS (T_C = 25 °C unless otherwise noted)

PARAMETER		SYMBOL	MBR2535CT	MBR2545CT	UNIT
Maximum repetitive peak reverse voltage		V_{RRM}	35	45	V
Working peak reverse voltage		V_{RWM}	35	45	V
Maximum DC blocking voltage		V_{DC}	35	45	V
Maximum average forward rectified current at T_C = 130 °C	total device	$I_{F(AV)}$	25		A
	per diode		12.5		A
Peak forward surge current 8.3 ms single half sine-wave superimposed on rated load per diode		I_{FSM}	150		A
Peak repetitive reverse surge current per diode at t_p = 2 µs, 1 kHz		I_{RRM}	1.0		A
Peak non-repetitive reverse energy (8/20 µs waveform) per diode		E_{RSM}	25		mJ
Electrostatic discharge capacitor voltage human body model: C = 100 pF, R = 1.5 kΩ		V_C	25		kV
Voltage rate of change (rated V_R)		dV/dt	10 000		V/µs
Operating junction temperature range		T_J	-65 to +150		°C
Storage temperature range		T_{STG}	-65 to +175		°C

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ELECTRICAL CHARACTERISTICS (T_C = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**MBR2535CT**	**MBR2545CT**	**UNIT**
Maximum instantaneous forward voltage per diode	I_F = 30 A	T_C = 25 °C	V_F (1)	0.82		V
		T_C = 125 °C		0.73		
Maximum instantaneous reverse current at blocking voltage per diode		T_C = 25 °C	I_R (1)	0.2		mA
		T_C = 125 °C		40		

Note

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

THERMAL CHARACTERISTICS (T_C = 25 °C unless otherwise noted)			
PARAMETER	**SYMBOL**	**MBR**	**UNIT**
Typical thermal resistance from junction to case per diode	$R_{\theta JC}$	1.5	°C/W

ORDERING INFORMATION (Example)					
PACKAGE	**PREFERRED P/N**	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
TO-220AB	MBR2545CT-E3/45	1.85	45	50/tube	Tube

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RATINGS AND CHARACTERISTICS CURVES (T_C = 25 °C unless otherwise noted)

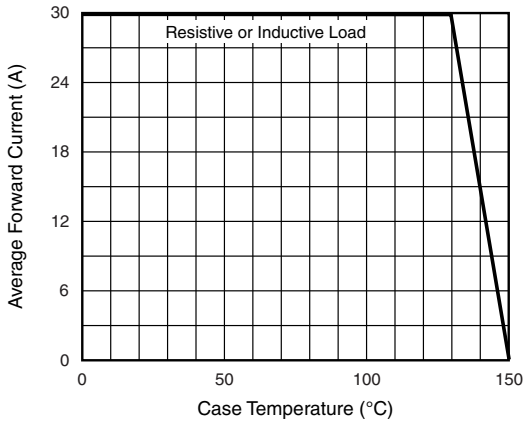


Fig. 1 - Forward Current Derating Curve

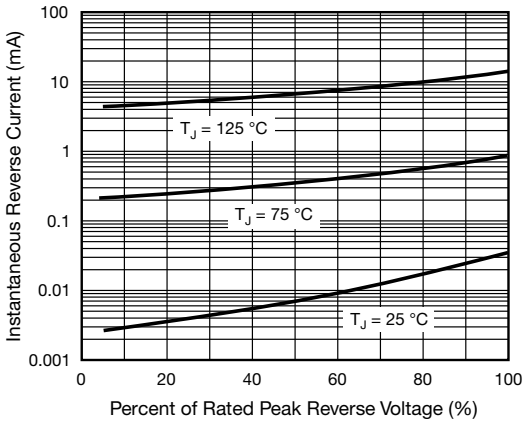


Fig. 4 - Typical Reverse Characteristics Per Diode

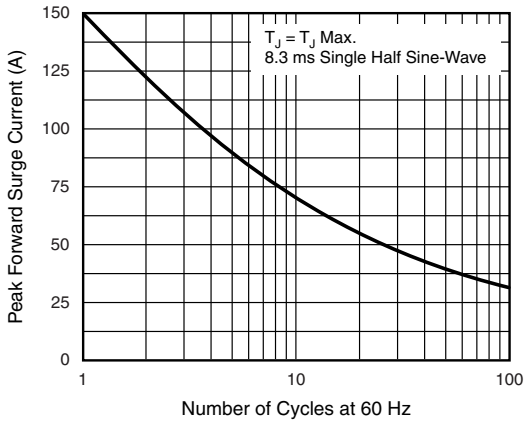


Fig. 2 - Maximum Non-Repetitive Peak Forward Surge Current Per Diode

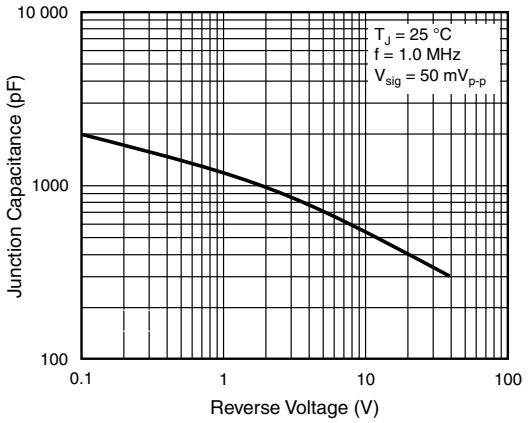


Fig. 5 - Typical Junction Capacitance Per Diode

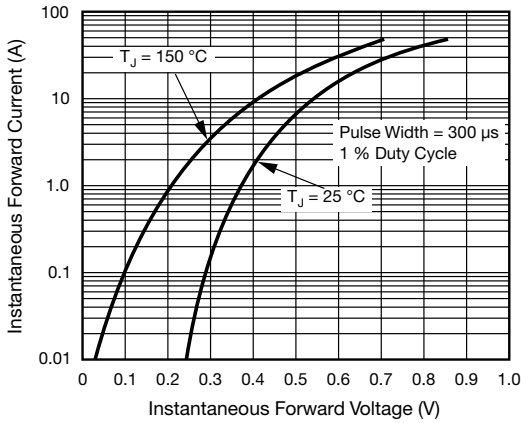


Fig. 3 - Typical Instantaneous Forward Characteristics Per Diode

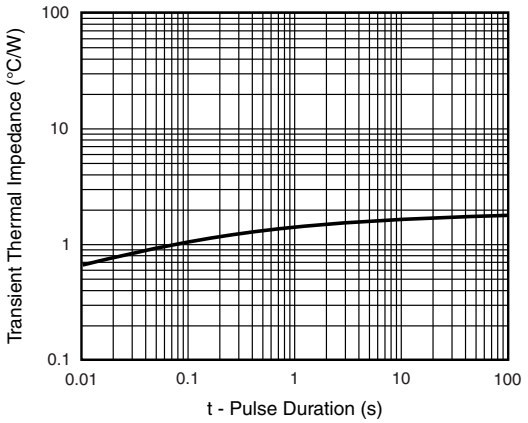
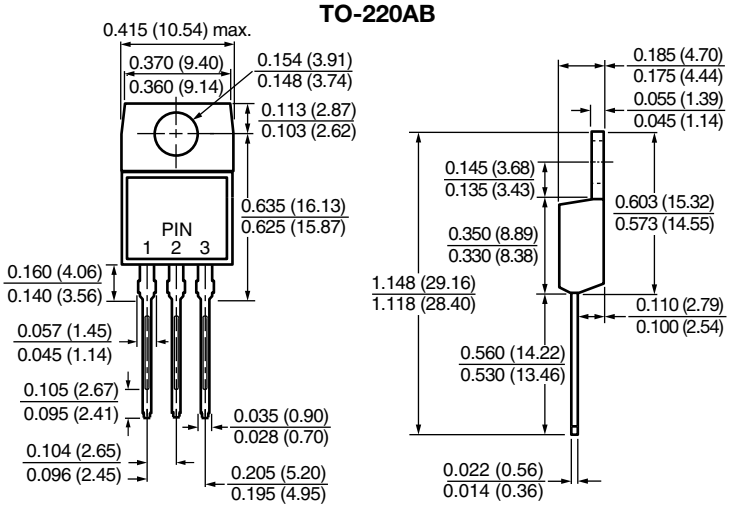


Fig. 6 - Typical Transient Thermal Impedance Per Diode

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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

TO-220AB



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